The Bono de Atracción and Teacher Retention in Peru: Regression Discontinuity Evidence

Working notes – first results, data inventory, and next steps

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Abstract

We study the effects of Peru's *Bono de Atracción*—an unconditional-on-location bonus of S/ 18,000 (about USD 4,800) paid over three years to teachers who enter the permanent public-school career (nombramiento) in the top third of their cohort's final merit ranking—on position take-up, retention, and early-career progression. We exploit the sharp discontinuity in bonus eligibility at the top-third cutoff of the final concurso score, pooling the 2015, 2017, 2018 and 2019 national hiring concursos (33,383 winners, of whom 11,147 received the bonus). Assignment is essentially deterministic: crossing the cutoff raises the probability of receiving the bonus from 0 to 0.992. Take-up of the permanent position is nearly universal on both sides of the cutoff (98.3% just below), leaving little room for effects on the extensive margin: the RD estimate on being on the public payroll the year after the concurso is +0.3 percentage points (pp) and statistically indistinguishable from zero. Retention point estimates are uniformly positive but small and imprecise (+0.6 pp at $t+2$, $p=0.20$; +0.7 pp at $t+4$), with the clearest signal in the 2015 cohort—the only one we can follow for six years (+1.5 pp at $t+2$, $p=0.06$). We find a precise zero on remaining in the school where the teacher was appointed, consistent with the rule that the bonus follows the teacher rather than the school. The aggregate zero masks offsetting responses: among winners assigned to *rural* schools, recipients are less likely to remain in their assigned school (-2.6 pp at $t+1$, $p=0.05$) and in the rural system altogether (-2.2 pp at $t+2$), while the aggregate retention gain is concentrated among urban appointees (+1.2 pp, $p=0.04$)—the windfall appears to finance the move out of hard-to-staff schools. These notes document the institutional rules, the data inventory, the identification checks, and a plan to extend the analysis to the 2022 and 2024 concursos and to richer outcomes; marked pointers throughout indicate where additional data would most strengthen the analysis.

*These notes were prepared with research assistance from Claude (Anthropic). All data work is documented in the accompanying code (`code/01`–`04`); nothing in the source data folders was modified. Comments and corrections are very welcome.

1 Motivation and summary

Attracting high-ability candidates into the teaching profession—and keeping them there—is a first-order concern for education systems in developing countries. A growing literature studies pay reforms and location-based incentives for teachers, but surprisingly little is known about *selective* financial incentives that reward entering the profession at the top of the ability distribution, holding the job and its location fixed. Peru's *Bono de Atracción* is exactly such a policy: teachers who enter the Carrera Pública Magisterial (CPM) in the top third of the final merit ranking of their national hiring concurso receive S/ 6,000 per year during their first three years of service, conditional on remaining active permanent teachers—anywhere in the public system.

These notes report a first pass at the question using a regression discontinuity (RD) design around the top-third cutoff. Our reading of the evidence so far is threefold. First, the design works mechanically very well: the running variable (the unbonified final concurso score) assigns the bonus essentially deterministically once the tie-broken mass point at the cutoff score is set aside, and standard validity checks (density, covariate balance) raise no red flags. Second, the margin the bonus was named for—attracting winners into actually taking the position—has virtually no room to move: about 98.3% of winners just below the cutoff appear on the public payroll as permanent teachers the following year. Third, the interesting action is in *where* teachers stay rather than *whether* they stay: same-school retention—the margin the bonus does *not* pay for—shows a precise zero on aggregate, but the zero masks offsetting responses. Rural appointees who receive the bonus become less likely to remain in their assigned school ($p = 0.05$) and in any rural school, without leaving the profession, while the aggregate retention gain (+1.2 pp at $t + 2$, $p = 0.04$) is entirely an urban-appointee phenomenon: the bonus appears to finance exit from exactly the schools where retention is scarce. Sections 2–5 document these results; Sections 6–7 lay out the data we still need and how the project could become a paper.

2 Institutional background

2.1 Entry into the permanent teaching career

Since the 2012 Ley de Reforma Magisterial (Ley 29944), entry into a permanent public-school teaching position (*nombramiento*) occurs exclusively through national concursos. Each concurso has two stages. In the national stage, all applicants take the Prueba Única Nacional (PUN), a centralized exam; those clearing minimum scores become *habilitados* for the decentralized stage, in which they apply to specific schools and are evaluated locally (professional background, a classroom observation, and an interview). The final score is the sum of the two stages, and winners (*ganadores*)—applicants ranked first for a specific vacancy—receive a nombramiento resolution and take up their plaza the following school year. Concursos with a bono round were held in 2015, 2017, 2018, 2019, 2022 and 2024; this draft uses the first four, for which we have complete rankings microdata (Table 1).

2.2 The Bono de Atracción

Article 22 of Ley 30281 (the 2015 budget law) created a bonus for teachers "who enter the Carrera Pública Magisterial occupying the top third of the final merit ranking of the public nombramiento concursos." The rules are laid out in RM 666-2018-MINEDU and, since April 2021, in RVM 118-2021-MINEDU. The features that matter for our design and interpretation are the following.

1. **Amount and timing.** S/ 6,000 per year for three consecutive years, starting the year after the concurso, paid each December through the regional payroll units. The total, S/ 18,000, is roughly seven months of the entry-level teacher salary. The bonus is neither pensionable nor remunerative.

2. **Eligibility.** The top third is computed *nationally*, across all levels and specialties, among concurso winners registered as *nombrado* in the NEXUS payroll system at a validation date, excluding teachers who were awarded the bonus in a previous concurso. The count of eligible winners is divided by three (rounded) and the bonus goes to the top of the ranking. Under the 2015–2019 rounds the ranking used the total (final) concurso score; under RVM 118-2021—which governs the 2022 and 2024 rounds—the ranking uses the PUN score only. In both regimes the ranking score *excludes* the percentage bonifications (disability, armed forces) that are added for plaza-allocation purposes.

3. **Conditionality.** Each annual payment requires (i) an active nombramiento, (ii) at least five months of effective classroom service during the school year (verified by the UGEL each September 15), and (iii) not having been removed from the CPM. Crucially, service may be *in any public school*—transfers (reasignaciones) do not void the bonus. Missed years are not recovered; resignation, dismissal or nullity of the appointment forfeits all remaining payments.

Point 3 shapes what we should expect: the bonus pays teachers to remain permanent public-school teachers during their first three years, but it does *not* pay them to remain in the (often rural, hard-to-staff) school where they were appointed. Effects on same-school retention would therefore have to operate through an income effect or through selection, not through the explicit incentive.

▷ **More data would help here:** *Two institutional details remain unverified and are worth an information request to MINEDU (DIBRED): the exact tie-breaking rule applied at the cutoff score, and the NEXUS validation dates used to define the eligible pool in each round. Both matter for the handful of observations at and just above the cutoff (Section 3).*

3 Data

3.1 Sources

We combine three administrative sources, all matched by national ID (DNI).

Table 1: Concurso rounds, winners, and the top-third cutoff

Concurso	Winners	Bono recipients	Share	Cutoff score	Payroll horizon
2015	8,137	2,702	33.2%	228.0	$t+1,\ldots,t+6$
2017	10,759	3,611	33.6%	227.5	$t+1,\ldots,t+4$
2018	9,975	3,348	33.6%	224.0	$t+1,\ldots,t+3$
2019	4,512	1,486	32.9%	222.0	$t+1, t+2$
2022	–	9,999[a]	–	–	(data plan, Section 6)
2024	–	8,210[a]	–	–	(data plan, Section 6)

Winners: ganadores with an adjudicated plaza in the cleaned concurso files. Bono recipients: rows in the digitized beneficiary lists matched to winners by DNI ($\geq$99.8% match). Cutoff: minimum unbonified final score among recipients. Payroll horizon: years relative to the concurso observable in the NEXUS 2015–2021 annual extraction.

[a] Beneficiary list counts; concurso rankings not yet digitized.

Bono beneficiary lists. MINEDU publishes the list of beneficiaries for each round; we digitized the PDFs into CSVs (2,702; 3,617; 3,351; 1,489; 9,999; and 8,210 names for 2015–2024). The 2015 list reports the final score and the national merit rank; the 2017 list reports both the concurso score (with bonifications) and the score used for the bono ranking; the 2018–2019 lists report the total score; the 2022 and 2024 lists report a score on a different scale, consistent with the PUN-only ranking rule in RVM 118-2021. Appendix A records our file-by-file reading of these lists and the open questions.

Concurso rankings. For 2015–2019 we have cleaned microdata covering all 288,757 registered applicants (PUN scores), the 86,807 habilitados, the 153,779 decentralized-stage records (including sub-scores, bonifications, and the assigned school), and the 33,383 winners. Winners merge to the bono lists by DNI at $\geq$99.8%.

NEXUS payroll. The Sistema de Administración y Control de Plazas records every teaching position in the country. We use an annual (Nov/Dec) national extraction for 2015–2021 (3.07 million position-year records) with the incumbent's DNI, appointment status (nombrado/contratado/designado), school code, position type, and pay scale (escala magisterial). This is the same system MINEDU itself uses to validate bono payments.

3.2 The running variable and the cutoff

For 2015–2019 the bono ranking score is the final concurso score *excluding* law-mandated percentage bonifications. In the winners file the recorded final score embeds those bonifications for about 2% of winners; we verified that for all 217 bono recipients whose recorded score differs from the score on the beneficiary list, the gap equals exactly the sum of the bonification sub-scores recorded in the decentralized-stage file. We therefore use the unbonified score throughout. With this correction, assignment is essentially deterministic: *no* recipient lies strictly below the round-specific cutoff (the minimum recipient score), and only 87 of 22,236 non-recipient winners (0.4%) lie strictly above it—consistent with the norm's exclusions (winners without a validated nombramiento at the cut-off

date, and prior beneficiaries).

One feature requires care: the cutoff score itself is a mass point where eligibility was resolved by a tie-breaking rule we do not observe (at the 2015 cutoff score, 56 of 81 winners received the bonus; at the 2018 cutoff score, only 4 of 99 did). Our baseline specification therefore drops the cutoff-score mass point (a "donut" of one score value); estimates including it, treating eligibility as fuzzy, are reported as robustness and are very similar.

3.3 Outcomes

For each winner we construct, from the payroll: (i) *acceptance*, an indicator for appearing as nombrado in the November/December payroll the year after the concurso; (ii) *retention*, the same indicator at $t+2, \ldots, t+6$; (iii) *same-school retention*, being nombrado in the school where the concurso assigned them; (iv) *career progression*, having advanced beyond the entry pay scale by 2021 and ever holding a director/directivo position. Winners absent from the payroll in a given year are coded zero for that year.

▷ **More data would help here:** *The binding data constraint for the whole project is the payroll horizon: the NEXUS extraction ends in 2021, which truncates the 2018–2019 cohorts at $t+3/t+2$ and rules out the 2022/2024 cohorts entirely (Section 6, item 4). Two NEXUS panel files in the Dropbox remain online-only placeholders (`nexus_panel_teacher_long_new.dta`, `nexus_panel_teacher_wide_new.dta`); they are not needed for the current results but should be synced before we rebuild anything that uses them. Finally, monthly (rather than Nov/Dec) NEXUS snapshots—which exist in the raw sources for some years—would let us measure within-year exits and the exact five-month service condition MINEDU verifies.*

4 Empirical strategy

We compare concurso winners just above and just below their round's top-third cutoff. Let r_{ic} denote winner i's unbonified final score centered at the cutoff of concurso c. We estimate, pooling rounds and using `rdrobust` (local linear, triangular kernel, MSE-optimal bandwidths, robust bias-corrected inference),

$$y_{ic} = \alpha + \tau\, \mathbf{1}\{r_{ic} \geq 0\} + f(r_{ic}) + \varepsilon_{ic}, \tag{1}$$

where $f(\cdot)$ is estimated separately on each side of the cutoff. Because assignment is deterministic in the donut sample, the reduced form and the fuzzy RD (instrumenting bonus receipt with cutoff crossing) coincide up to the 0.4% of protocol exclusions; we report both.

Two features make the design unusually credible. First, the cutoff is determined *ex post* by the realized distribution of scores—the number of winners divided by three—so applicants could not target an unknown threshold even if they could manipulate scores. Second, everyone in the sample already won a permanent position, so both sides of the cutoff face identical job offers; the only discontinuity is the S/ 18,000. Consistent with this, the density of the running variable is smooth

through the cutoff (pooled $p = 0.43$) and predetermined covariates (sex, age, rurality of the assigned school, baseline payroll status) are balanced (Table 6, Figure 7).

5 Results

5.1 First stage

Figure 1 shows the share of winners on the beneficiary list by half-point score bins, by round and pooled. The probability of receiving the bonus jumps from 0 to 0.992 at the cutoff (local linear, ±10-point bandwidth, donut sample; Table 2). The first stage is, for all practical purposes, one.

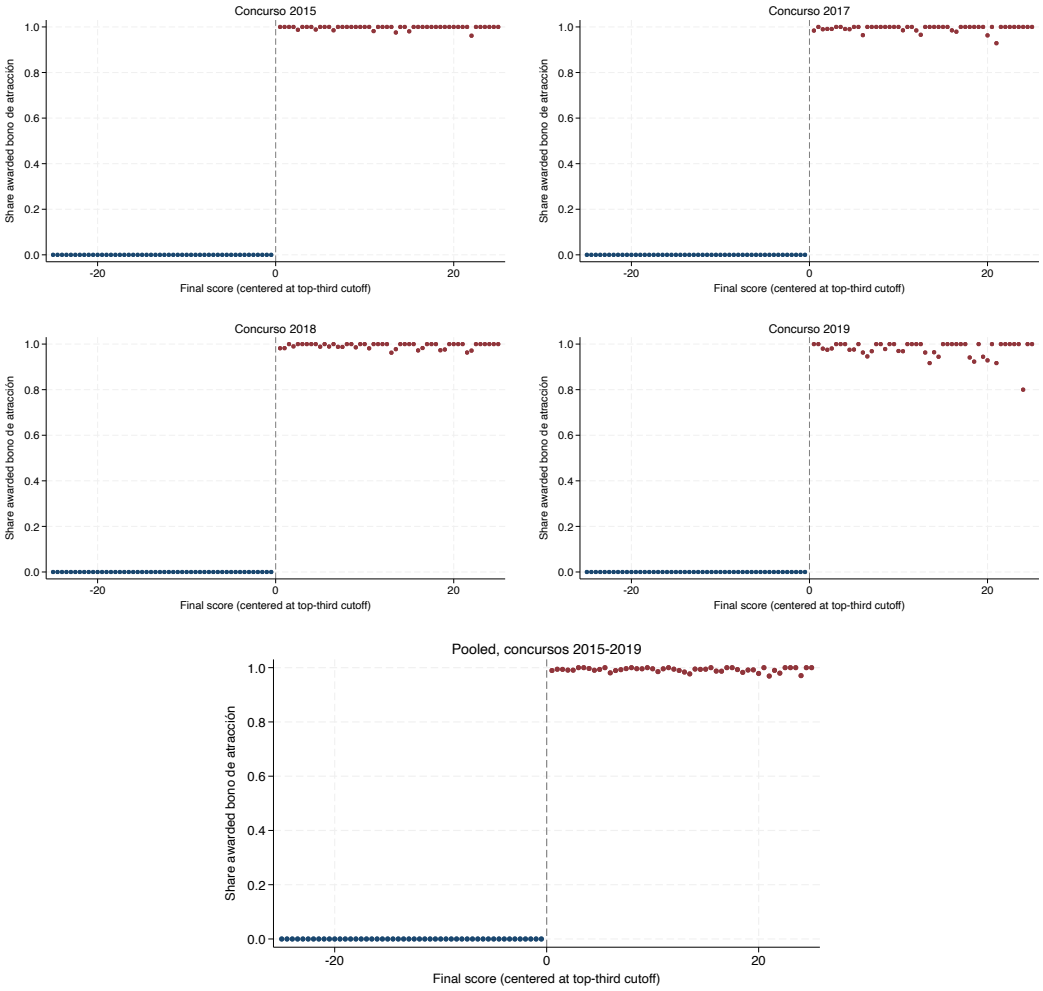


Figure 1: First stage: probability of receiving the Bono de Atracción. Half-point bins of the unbonified final score, centered at the round-specific top-third cutoff; the tie-broken cutoff-score mass point is excluded.

Table 2: First-stage estimates (donut sample)

	2015	2017	2018	2019	Pooled
Bandwidth ±5 *points*					
RD estimate	1.000	0.990	0.986	0.994	0.992
	(0.002)	(0.006)	(0.008)	(0.010)	(0.003)
Obs. left/right	984/823	1261/1107	1095/942	542/444	3882/3316
Bandwidth ±10 *points*					
RD estimate	0.997	0.990	0.991	0.991	0.992
	(0.002)	(0.004)	(0.004)	(0.007)	(0.002)
Obs. left/right	2016/1503	2575/2013	2281/1746	1092/792	7964/6054

Local linear regression of bonus receipt on an above-cutoff indicator, the centered score, and their interaction, within the indicated bandwidth, excluding the cutoff-score mass point. Robust standard errors in parentheses.

5.2 Acceptance and retention

Take-up of the position is close to universal on both sides of the cutoff: 98.3% of winners in the five points below the cutoff are nombrados on the public payroll the following November. The RD estimate is +0.3 pp (robust $p = 0.55$; Table 3, Figure 2). Permanent positions in Peru are valuable enough that essentially every winner takes one; there is simply no margin for the bonus to move here. The same holds for being on the payroll under any status.

Retention estimates are uniformly positive but small and individually insignificant: +0.6 pp at $t+2$ ($p = 0.20$) and +0.7 pp at $t+4$ ($p = 0.35$), against a control mean around 98%. The dynamics are displayed in Figure 3. The clearest signal comes from the 2015 cohort, the only one we can follow six years out: +1.5 pp at $t + 2$ ($p = 0.06$) and +1.2 pp at $t + 6$ ($p = 0.36$). With attrition from the CPM this low, even a bonus of seven months' salary can at most move retention by a point or two—the ceiling is the story. Still, two readings of the positive pattern remain open, and we cannot yet separate them: a genuine retention effect concentrated among the few would-be leavers, or noise. The horizon extension to 2022–2025 payroll data (Section 6) will help, both by adding the 2018–2019 cohorts' post-payment years (if the effect is incentive-driven, it should plateau or fade once payments stop) and by adding statistical power.

Same-school retention shows no discontinuity at any horizon (-0.5 pp at $t + 1$, SE 0.7, on a base of 97.0%): recipients are no more likely to stay in the school that appointed them. This is exactly what the rules predict—the bonus follows the teacher, not the school—and it suggests the policy purchases system-level retention (already near its ceiling) rather than retention where scarcity actually binds. We consider this one of the most policy-relevant facts so far, and Section 5.4 pushes on it by splitting the sample by the rurality of the assigned school.

▷ **More data would help here:** *Extending the payroll to 2022–2025 does double duty in this section: it adds the post-payment years for the 2018–2019 cohorts (the test for whether the retention effect fades when the money stops) and it roughly doubles the pooled sample at horizons $t + 3$ and beyond, where the point*

estimates are largest but the confidence intervals are widest.

5.3 Career progression

Neither career outcome moves robustly. Marginal recipients are +1.0 pp more likely to ever hold a director/directivo position by 2021 ($p = 0.31$), on a control mean of 14.0%; the estimate was larger and marginally significant before excluding the tie mass point (+2.6 pp, $p = 0.06$, fuzzy), so we flag it as worth revisiting with the directivo concurso files (which we have) rather than payroll job codes alone. Progression beyond the entry pay scale by 2021 shows a noisy negative estimate (-2.7 pp, $p = 0.26$, 2015/2017 cohorts), which we do not over-interpret; ascenso opportunities were limited and cohort-specific.

▷ **More data would help here:** *Career outcomes are currently inferred from payroll job codes and pay scales alone. The ascenso concurso results (2018, 2019, 2021) and the directivo concurso files (habilitados and ganadores, already in* `concursos/clean`*) would let us measure* attempted *versus* achieved *promotion separately—the margin an income windfall should move first. Administrative salary data (SUP/planillas) would also let us put a soles value on each career transition and calibrate the income effect.*

Table 3: RD estimates: acceptance, retention, and career outcomes (pooled)

	Reduced form		Fuzzy RD			Control
Outcome	Estimate	(SE)	Estimate	(SE)	N (bw)	mean
Nombrado in payroll, $t+1$ (acceptance)	0.003	(0.004)	0.003	(0.004)	18,694	0.983
Nombrado, $t+2$	0.006	(0.004)	0.006	(0.004)	16,388	0.983
Nombrado, $t+3$	0.006	(0.004)	0.006	(0.004)	15,223	0.983
Nombrado, $t+4$	0.007	(0.007)	0.007	(0.007)	7,758	0.981
In payroll (any status), $t+1$	-0.000	(0.004)	-0.000	(0.004)	13,398	0.990
In payroll (any status), $t+2$	0.002	(0.004)	0.002	(0.004)	12,739	0.991
In payroll (any status), $t+3$	0.002	(0.004)	0.002	(0.004)	12,134	0.990
Nombrado in assigned school, $t+1$	-0.005	(0.007)	-0.005	(0.007)	13,398	0.970
Nombrado in assigned school, $t+2$	-0.001	(0.007)	-0.001	(0.007)	17,010	0.961
Nombrado in assigned school, $t+3$	-0.003	(0.009)	-0.003	(0.009)	13,190	0.950
Ever director/directivo by 2021	0.010	(0.011)	0.010	(0.011)	18,694	0.140
Escala 2+ by 2021 (2015/17 cohorts)	-0.027	(0.024)	-0.027	(0.024)	8,450	0.395

Donut sample (cutoff-score mass point excluded). `rdrobust` local linear estimates, triangular kernel, MSE-optimal bandwidth; robust bias-corrected p-values: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Control mean: outcome mean for winners 0–5 points below the cutoff. Fuzzy RD instruments bonus receipt with cutoff crossing. Horizons $t+3$ ($t+4$) exclude the 2019 (2018–19) cohorts.

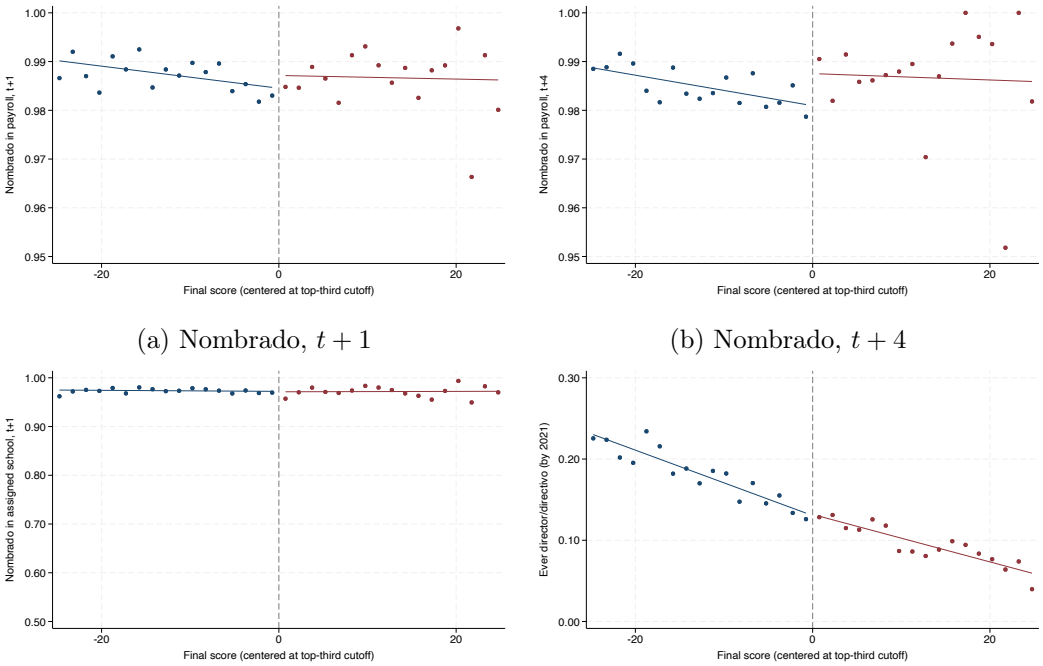
5.4 Heterogeneity

Table 5 splits the donut sample along the rurality and hardship classification of the assigned plaza—taken from the concurso plaza database, which records the allowance-relevant categories (urban, rural 1–3, bilingual, border, VRAEM) for every vacancy in all four rounds—plus sex, age, baseline employment, and teaching level. Figure 4 shows the rural/urban split for same-school retention, and

Table 4: RD estimates by cohort (donut sample, reduced form)

	2015	2017	2018	2019
Nombrado, $t+1$	0.014	0.001	-0.008	0.016
	(0.011)	(0.008)	(0.009)	(0.010)
Nombrado, $t+2$	0.015*	0.002	0.000	0.014
	(0.008)	(0.008)	(0.008)	(0.010)
Nombrado, $t+4$	0.012	0.002	–	–
	(0.009)	(0.008)	–	–
Nombrado, $t+5$	0.007	–	–	–
	(0.011)	–	–	–
Nombrado, $t+6$	0.012	–	–	–
	(0.012)	–	–	–

`rdrobust` local linear estimates by concurso; standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (robust bias-corrected). Later horizons only observable for earlier cohorts given payroll through 2021.



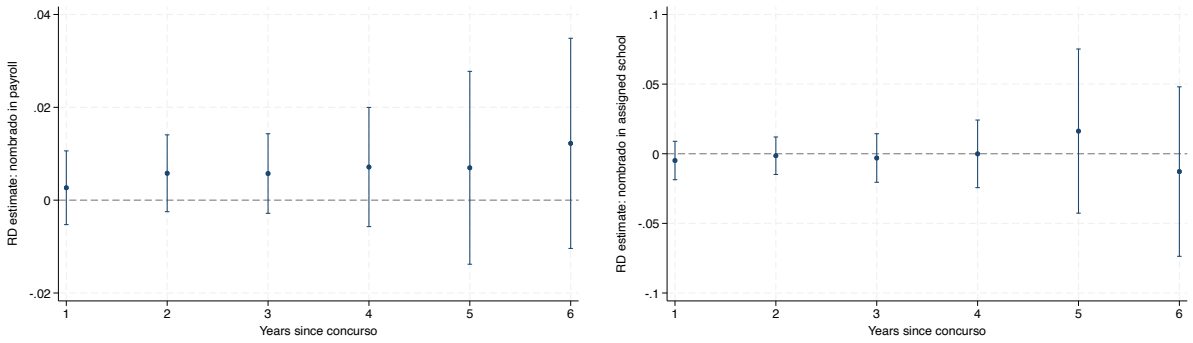
(a) Nombrado, $t+1$

(b) Nombrado, $t+4$

(c) Same school, $t+1$

(d) Ever director by 2021

Figure 2: Reduced-form discontinuities, pooled concursos. Bins of 1.5 score points; linear fits weighted by bin size on each side.



(a) Nombrado in payroll

(b) Nombrado in assigned school

Figure 3: RD estimates by years since the concurso (pooled; later horizons identified by earlier cohorts only). Bars are 95% confidence intervals.

Figure 5 the key comparison within rural appointees. The last column adds a new outcome built for this section: teaching (as nombrado) in *any* rural school at $t + 2$, using a school-level rurality lookup constructed from the same plaza database.

The rural/urban contrast is the headline, and with all four cohorts it sharpens considerably. Among winners assigned to *rural* schools, bonus recipients are less likely to still be in their assigned school one year later (-2.6 pp, $p = 0.05$; -2.1 pp at $t+2$, $p = 0.09$), while the urban estimate is a small positive (+0.5 pp at $t+1$); the rural–urban difference at $t+1$ is -3.1 pp (SE 1.5, t = -2.1). Crucially, the new rural-system outcome shows that the movers do not relocate to other rural schools: among rural appointees, the probability of teaching in *any* rural school at $t + 2$ falls by -2.2 pp ($p = 0.08$, against a control mean of 96.3%), while their system-level retention is exactly flat (-0.1 pp at $t+2$). The exit is largest where exit is cheapest—Rural 1 schools, the least remote category (-5.1 pp at $t+1$, $p = 0.03$)—and the degree gradient is not monotone beyond that. Taken together, the pattern reads as urban drift financed by the windfall: because the bonus follows the teacher anywhere in the public system, the S/ 6,000 payments can cover exactly the costs (paperwork, travel, temporary double residence) of moving out of the hard-to-staff schools the concurso placed these teachers in.

The mirror image is that the aggregate retention effect is entirely an urban-appointee phenomenon: system retention at $t + 2$ rises by +1.2 pp among urban appointees ($p = 0.04$) and not at all among rural ones. As before, the retention effect is also concentrated among primary-level teachers (+1.6 pp at $t + 2$, $p = 0.03$) and, more weakly, among women (+0.8 pp, $p = 0.09$); there is no heterogeneity by age or by prior payroll status. The bilingual, VRAEM and border subsamples are too small for informative estimates at the cutoff (their confidence intervals span economically large effects of either sign). Two caveats before we lean on any of this: the subgroup count is now large enough that a couple of 5–10% p-values are expected under the null, and the rural/urban split—though predetermined—was chosen after seeing the aggregate same-school zero. The right discipline is a pre-specified follow-up: the 2022/2024 cohorts (Section 6) provide out-of-sample replication material once their payroll years accumulate.

Table 5: Heterogeneous RD effects (donut sample, reduced form)

	Nombrado $t+1$	Nombrado $t+2$	Same school $t+1$	Same school $t+2$	Rural school $t+2$
All winners	0.003	0.006	-0.005	-0.001	0.019
	(0.004)	(0.004)	(0.007)	(0.007)	(0.017)
Rurality of assigned school					
Rural	-0.001	-0.001	-0.026**	-0.021*	-0.022*
	(0.007)	(0.006)	(0.013)	(0.013)	(0.012)
Urban	0.005	0.012**	0.005	0.014	0.001
	(0.006)	(0.006)	(0.008)	(0.009)	(0.002)
Degree of rurality					
Rural 1 (least remote)	-0.008	-0.009	-0.051**	-0.036	-0.023
	(0.016)	(0.016)	(0.024)	(0.024)	(0.020)
Rural 2	0.012	0.006	-0.001	-0.019	-0.026
	(0.011)	(0.011)	(0.020)	(0.026)	(0.025)
Rural 3 (most remote)	-0.015	-0.010	-0.035	-0.022	-0.028
	(0.011)	(0.010)	(0.020)	(0.024)	(0.023)
Other hardship categories					
Bilingual school	-0.003	-0.000	0.016	0.022	-0.046
	(0.017)	(0.020)	(0.024)	(0.037)	(0.084)
VRAEM school	-0.059	-0.082	-0.050	-0.051	-0.170
	(0.048)	(0.073)	(0.033)	(0.047)	(0.118)
Border school	0.008	0.009	0.018	0.015	0.150*
	(0.009)	(0.009)	(0.020)	(0.023)	(0.076)
Sex					
Female	0.004	0.008*	-0.005	0.001	0.023
	(0.005)	(0.004)	(0.009)	(0.008)	(0.022)
Male	-0.002	0.002	-0.004	-0.006	0.016
	(0.008)	(0.009)	(0.011)	(0.012)	(0.027)
Age					
Below median age	0.005	0.009	-0.002	0.000	0.025
	(0.006)	(0.006)	(0.010)	(0.011)	(0.024)
Above median age	0.001	0.003	-0.006	-0.003	0.016
	(0.006)	(0.006)	(0.008)	(0.009)	(0.025)
Baseline employment					
In payroll at t_0	0.004	0.006	-0.007	-0.003	0.014
	(0.005)	(0.005)	(0.008)	(0.009)	(0.019)
New entrant at t_0	-0.001	0.008	0.004	0.013	0.064
	(0.016)	(0.015)	(0.019)	(0.019)	(0.045)
Teaching level					
Inicial	0.003	0.003	-0.000	-0.006	-0.014
	(0.007)	(0.005)	(0.009)	(0.010)	(0.041)
Primaria	0.012*	0.016**	-0.009	0.005	0.010
	(0.007)	(0.007)	(0.013)	(0.013)	(0.033)
Secundaria	-0.002	0.002	-0.002	-0.003	0.039
	(0.009)	(0.008)	(0.011)	(0.010)	(0.027)

`rdrobust` local linear estimates by subgroup; standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (robust bias-corrected). Rurality and hardship categories of the assigned plaza come from the concurso plaza database (all four rounds, 100% coverage). "Rural school $t+2$" is teaching as nombrado in any rural school two years after the concurso, using a school-level rurality lookup from the same database. Age split at the sample median (35). Baseline employment measured in the November payroll of the concurso year.

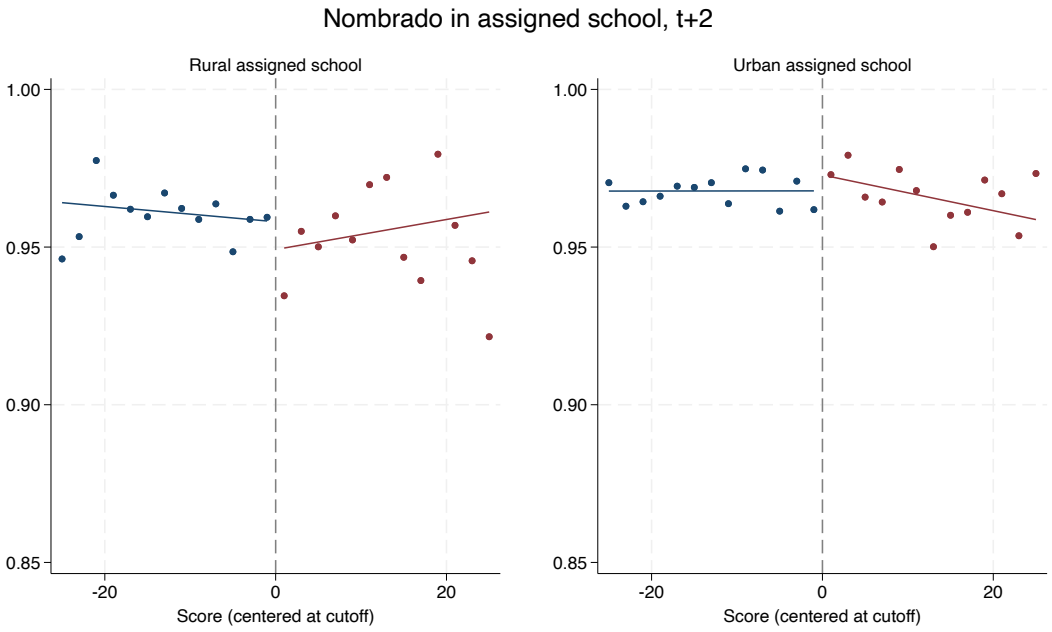


Figure 4: Same-school retention at $t + 2$ by rurality of the assigned school (2-point bins, donut sample, all four concursos).

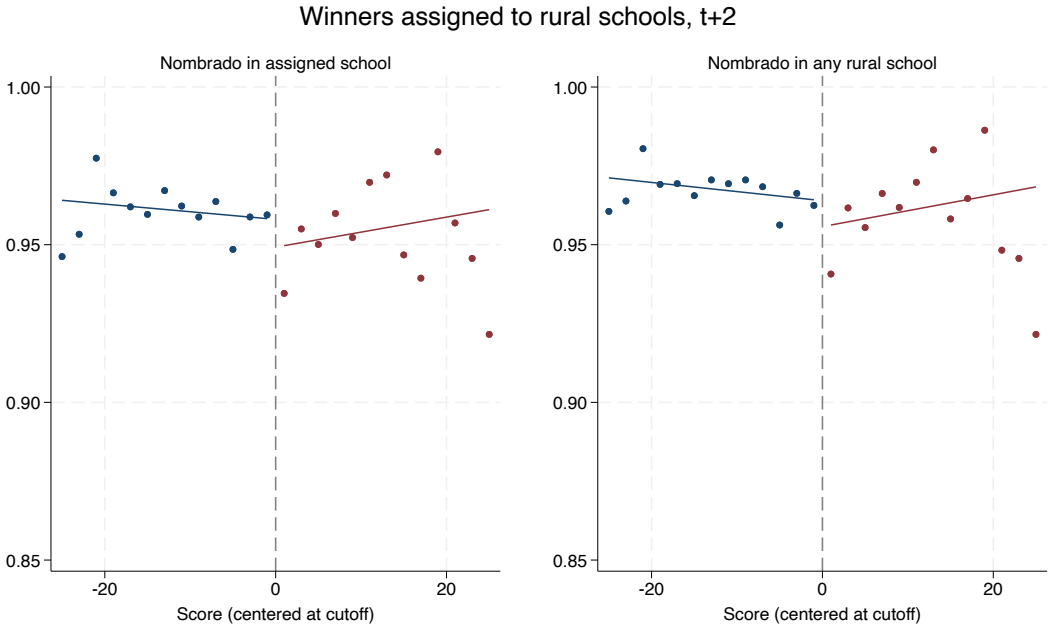


Figure 5: Winners assigned to rural schools: retention in the assigned school versus in any rural school, $t + 2$ (2-point bins, donut sample).

▷ **More data would help here:** *The rurality gap flagged in the previous draft is now closed (the plaza database covers all cohorts and adds bilingual/border/VRAEM), and the new any-rural-school outcome already tells us the movers leave the rural system. What would still sharpen this section: (i) reasignación (transfer) records, to see* where *movers go (nearby towns versus across regions) and* when *within the three payment years; (ii) a continuous hardship measure (distance to the provincial capital, from school coordinates in the padrón) to replace the four-category classification; (iii) a school-level rurality* panel *(the padrones de asignaciones, 2014–2026, are in the Dropbox but not yet synced offline)—our current lookup assigns each school its latest classification, so reclassifications over time add noise to the any-rural outcome.*

5.5 Validity and robustness

Table 6 reports manipulation tests and covariate balance; Figure 7 shows the corresponding balance plots, and Figure 6 the raw density. Covariates are smooth through the cutoff, and the pooled density test does not reject ($p = 0.43$). The 2015 round alone rejects at the 5% level ($p = 0.03$); we read this as an artifact of the discrete score grid (scores move in half-point steps, and `rddensity` is built for continuous running variables) rather than manipulation—the cutoff is a function of the entire realized score distribution and was unknowable at the time of scoring—but we will implement discrete-running-variable diagnostics (e.g., Frandsen's test) in the next iteration. Table 7 shows that the headline estimates are stable when we include the tie mass point (fuzzy), halve or double the bandwidth, or use a local quadratic.

Table 6: Validity: density tests and covariate balance

	Statistic / RD estimate	p-value
Density tests (rddensity)		
Concurso 2015	-2.23	0.03
Concurso 2017	0.13	0.89
Concurso 2018	-0.57	0.57
Concurso 2019	-0.83	0.41
Concurso Pooled	-0.78	0.43
Covariate balance (rdrobust, donut)		
Female	0.003	0.83
Age at concurso	0.179	0.50
Assigned school rural	0.025	0.29
Contract teacher, concurso year	-0.015	0.35
In payroll, concurso year	-0.015	0.25

Density: `rddensity` robust bias-corrected test. Balance: `rdrobust` estimates on predetermined covariates, donut sample.

Table 7: Robustness of the main estimates

	Nombrado $t+1$	Nombrado $t+2$	Same school $t+1$
Baseline (donut, MSE-optimal bw)	0.003	0.006	-0.005
	(0.004)	(0.004)	(0.007)
Including ties, fuzzy RD	0.005	0.008	-0.000
	(0.005)	(0.005)	(0.006)
Half bandwidth	0.003	0.006	-0.008
	(0.006)	(0.007)	(0.011)
Double bandwidth	0.002	0.004	-0.001
	(0.003)	(0.003)	(0.005)
Local quadratic	0.004	0.006	-0.007
	(0.006)	(0.006)	(0.009)

Each cell is a separate `rdrobust` estimate; standard errors in parentheses; * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (robust bias-corrected).

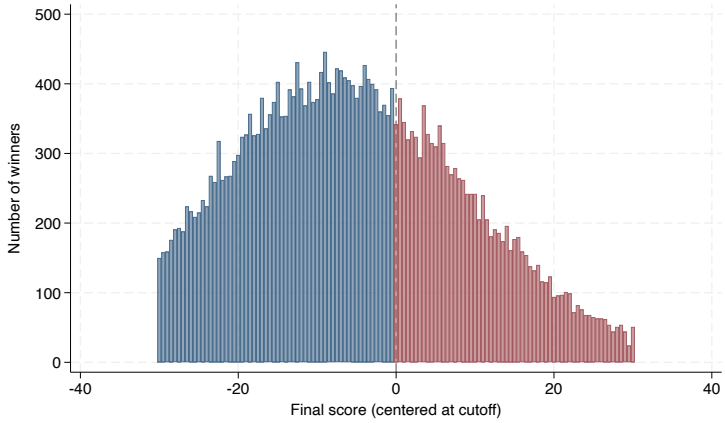


Figure 6: Density of the running variable (pooled), half-point bins.

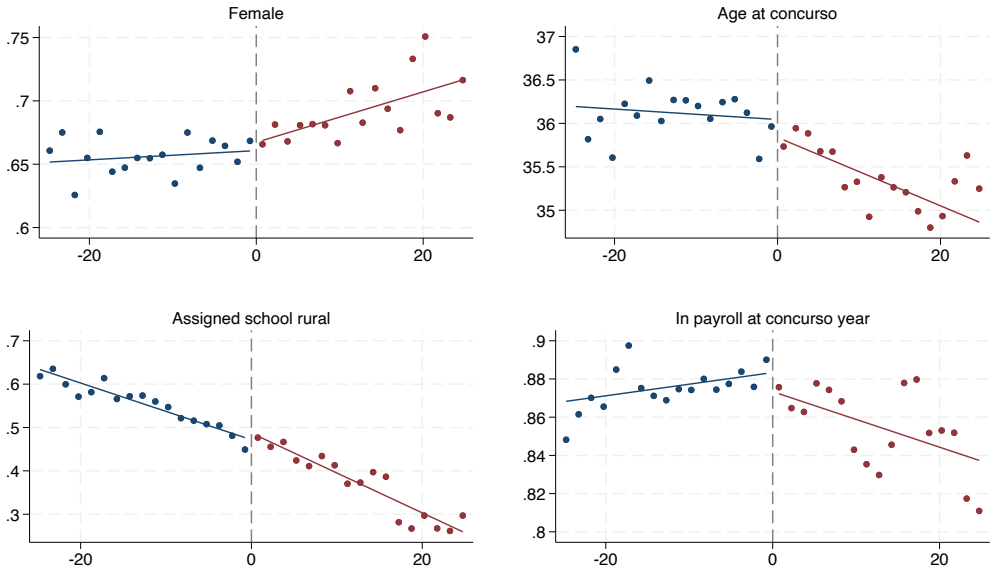


Figure 7: Covariate balance at the cutoff (pooled, donut sample).

6 Data acquisition plan: 2022 and 2024 concursos (and beyond)

Everything missing is public on MINEDU's evaluation portal (`https://evaluaciondocente.perueduca.pe/concursos-y-evaluaciones/evaluaciones-anteriores/`), organized per concurso with the same architecture as the rounds we already digitized. We verified the relevant pages for 2022 (`/nombramiento2022/`) and 2024 (`/nombramiento24/`): final PUN results, decentralized-stage results, and winners lists are posted as per-region PDFs (26 files per set, same layout family as the 2015–2019 sources in `concursos/source`). The plan, executable by Claude in a future session:

1. **Download** (1–2 hours, scripted): crawl the results pages for nombramiento 2022 and 2024 (PUN results, decentralized-stage final results, ganadores; plus the 2021 round for completeness even though it had no bono) and store the PDFs under `concursos/source/` following the existing convention (`PUN/2022`, `N_descentralizada/2022`, `N_ganadores/2022`, etc.), leaving current files untouched.

2. **Digitize**: extract the tables to CSV with the same pdf-extraction pipeline used for the bono lists (the PDFs are digital, not scans; extraction is reliable and verifiable against row counts printed on the documents).

3. **Integrate**: append to the cleaned stacked files with `concurso==2022/2024` labels, matching the schemas of `ganadores_nombramiento.dta` and `etapa_desc_nombramiento.dta`, with a validation notebook comparing counts to MINEDU's published totals.

4. **Payroll extension**: the 2015–2021 NEXUS extraction must be extended to 2022–2025 to measure outcomes for the new cohorts and the post-bonus years of the 2018–2019 cohorts.

Options: (i) the SINAD information-request route already used for the DIGEDD extraction (the 2025 deliveries in `nexus/source` suggest requests are being answered); (ii) MINEDU's open-data portal; (iii) scraping the public NEXUS consultation tool. We recommend (i), mirroring the exact variable list of the 2015–2021 extraction.

For the 2022/2024 rounds the running variable is the PUN score (per RVM 118-2021), which is attractive: the PUN is machine-graded, so local manipulation is even less plausible, and the design extends to a second incentive regime.

7 Where this could go: ideas for the paper

1. **Retention where it binds.** The bonus follows the teacher, not the school, and the full-sample heterogeneity (Section 5.4) now shows the consequence directly: rural appointees who receive the bonus leave their assigned school and the rural system while staying in the profession, and the aggregate retention gain accrues entirely among urban appointees. The natural paper frames the bono as a test of *unconditional* versus *location-conditioned* teacher pay: an identical transfer, delivered without a location string, buys system retention where it is already high and finances mobility away from where retention binds. Next steps: transfer (reasignación) destinations and timing, a continuous hardship gradient, and out-of-sample replication in the 2022/2024 cohorts.

2. **Incentive dynamics.** Payments stop after year 3. With payroll through 2025, we can test for exit spikes when payments end (2015 cohort: 2019; 2017 cohort: 2021; 2018 cohort: 2022)—an attractive test separating price effects from selection.

3. **Career ladders.** The directivo result (+1.0 pp) can be validated with the directores concurso files (habilitados/ganadores, which we have for 2016–2021) and ascenso participation: does a windfall early in the career finance investment in promotion?

4. **Student outcomes.** School-level ECE test scores and dropout (available through 2018/2019) can be linked through the school of service: does retaining marginal top-third teachers move school outcomes? Power is a concern given the small retention effect; worth a back-of-the-envelope before investing.

5. **The attraction margin.** The RD identifies retention effects, not attraction. The 2015 introduction (announced before the concurso) and the 2021 rule change (final score → PUN) are candidate variation for whether the bonus changed who applies and how they prepare—e.g., a difference-in-differences on application volumes and applicant quality around 2015, or bunching-style tests on effort in the decentralized stage once the ranking stopped depending on it in 2021.

6. **Comparison with location-based bonuses.** Peru also pays permanent rural/frontier/VRAEM allowances. Contrasting the (unconditional on location) atracción bonus with location-conditioned

pay in the same system speaks directly to the design of teacher compensation in developing countries.

8 Relation to Bobba et al. (2026) and Ederer (2026), and two concrete extensions

Two recent papers study teacher compensation and sorting in exactly our setting—the Peruvian centralized teacher labor market—using the same administrative infrastructure (concurso rankings, NEXUS): Bobba, Ederer, León-Ciliotta, Neilson and Nieddu (2026; henceforth BELNN), on *location-based* pay, and Ederer (2026), on the *dynamics* of teacher–school matching. Neither examines the Bono de Atracción, and the three-way comparison is unusually clean because the three instruments live in the same institutional system and are of comparable size.

BELNN: person-based versus school-based pay. BELNN show that wage rigidity makes teachers sort on non-wage amenities, concentrating competent teachers in urban schools, and use an RD on the rural wage bonus (a ∼13% permanent wage increase at remoteness/population cutoffs) to show that *school-based* pay is powerful on the recruitment margin: new recruits' competency rises by 0.42σ and student math scores by 0.27σ. Their structural model prices the urban–rural amenity gap: the most remote schools would need roughly **S/ 500 per month** to compete with the average urban school. The Bono de Atracción pays exactly S/ 500 per month (S/ 6,000 per year) for three years—but attached to the *person*, not the school. Our RD delivers the missing empirical moment: the same money, stripped of the location string, buys essentially nothing on any staffing margin that matters for inequality (no acceptance effect, no rural retention, negative same-school retention among rural appointees). Two further points of contact. First, in a static sorting model with additively separable utility, a person-based transfer that follows the teacher anywhere does not change *relative* school utilities and should produce exactly zero sorting effects—our aggregate same-school zero confirms this prediction out of sample; the *negative* rural effect goes beyond it and points to a mobility-cost/liquidity channel absent from the baseline model (cash relaxes the fixed cost of executing a transfer). Second, BELNN find that even the *school-based* bonus leaves teacher retention unchanged at their cutoffs (their Table C.7); together with our results, neither instrument retains teachers in rural schools—location pay works through recruitment, and person pay works through neither.

Ederer: cash and the job ladder. Ederer's dynamic matching model rationalizes why urban–rural gaps in teacher effectiveness *widen* with experience: rural placements are the bottom of a job ladder, and teachers rematch upward over their careers. His headline counterfactual shows that *mandating* minimum spells in rural schools backfires through adverse selection at recruitment, and that negating it would require roughly S/ 282 per month (≈19% of base pay). Our heterogeneity results are, to our knowledge, the first quasi-experimental evidence on the ladder mechanism itself: an unconditional windfall *accelerates* the climb (rural appointees leave the assigned school and the rural system while staying in the profession), and the effect is largest exactly where the ladder is

cheapest to climb (Rural 1 schools). The rural/urban asymmetry in our estimates is the model's signature: for urban appointees—already near the top of the ladder—the only margin left is leaving the profession, and there the bonus *does* retain (+1.2 pp); for rural appointees the rematching margin dominates and system retention is flat. The quantitative coincidence is worth emphasizing: Peru already spends, per recipient, about *twice* the monthly amount Ederer estimates is needed to offset the cost of staying in a rural school—it just does not condition the payment on staying.

Extension A (of BELNN): the counterfactual reallocation of the bono budget. The bono rounds 2015–2019 disbursed roughly S/ 200 million (11,147 recipients × S/ 18,000). BELNN's wage-setting machinery (their Proposition 1) can price what that budget would buy if converted into their cost-minimizing, school-based schedule: how much of the rural effectiveness gap closes when the atracción budget is reallocated from person-based to location-based pay, holding total cost fixed? Our RD estimates discipline the model twice—as out-of-sample validation moments (the zero sorting response to person-based pay; the mobility-cost channel) and as the empirical description of the status quo policy being replaced. This is the natural next paper for the BELNN team: same model, same data, a real S/ 200M policy, and a sharp design.

Extension B (of Ederer): pricing a voluntary retention contract. Ederer evaluates a *mandate*; the bono suggests its market-based counterpart: condition the three S/ 6,000 installments on remaining in the assigned (rural) school, i.e., a voluntary dynamic retention contract rather than a lock-in. Because participation at each payment date is chosen ex post, the adverse-selection cost of the mandate is partially avoided, and the existing budget is approximately the right order of magnitude (S/ 500 > S/ 282 per month). Our RD moments—in particular the negative same-school and any-rural effects and their timing relative to the payment window—identify the income effect on mobility that his model needs to simulate this contract credibly. The policy question the extension answers: how much rural retention (and downstream student achievement) does the *same* S/ 200M buy under (i) the current unconditional design, (ii) a stay-in-school-conditioned design, and (iii) BELNN's recruitment-margin schedule?

For the standalone RD paper, this positioning also settles the framing: the Bono de Atracción is the empirical test of the *person-based* corner of the teacher compensation design space, completing a triptych whose other corners—school-based pay (BELNN) and mobility restrictions (Ederer)—are already written.

A Bono beneficiary lists: file-by-file inventory

- `bono_atraccion_2015.csv` (2,702 rows). Columns: names, DNI, *puntaje final*, region, *orden de mérito*. Header of the source PDF states the top-third rule over final scores among winners with a nombramiento resolution and adjudicated plaza. Fully consistent with the 2015 ganadores file (100% DNI match).

- `bono_atraccion_2017.csv` (3,617 rows). Columns include *puntaje concurso* and *puntaje bono*; they differ for 58 rows, and where they differ the concurso score equals the bono score

times a bonification factor (1.10/1.15/1.25). Best guess (verified against decentralized-stage bonification sub-scores): the bono ranking uses the unbonified score. Six DNIs on the list do not appear in our ganadores file; unresolved, plausibly late plaza adjudications.

- `bono_atraccion_2018.csv` (3,351 rows) and `bono_atraccion_2019.csv` (1,489 rows). Single *puntaje total* column; three DNIs each unmatched to ganadores. At the 2018 cutoff score only 4 of 99 winners received the bonus—the strongest evidence that ties at the cutoff were broken by an unobserved criterion (the norm suggests the order of the underlying ranking components; we drop the mass point).

- `bono_atraccion_2022.csv` (9,999 rows) and `bono_atraccion_2024.csv` (8,210 rows). Single *puntaje* column on a ∼150–190 scale, consistent with the PUN-only ranking of RVM 118-2021; includes a `dni_flag` for rows where the DNI failed a checksum in digitization (to re-verify against source PDFs). Not yet usable for RD without the 2022/2024 concurso rankings (Section 6).

- **Open questions for MINEDU/colleagues:** (i) the exact tie-breaking rule at the cutoff score; (ii) whether the 87 above-cutoff non-recipients are indeed protocol exclusions (no validated nombramiento at the cut date; prior beneficiaries); (iii) for 2022/2024, the exact validation date defining the eligible pool.

B Code and file map

All code lives in `Claude/Principals/Bono de atraccion/`: `code/01_build_rd_dataset.do` (RD base), `code/02_build_outcomes.do` (NEXUS outcomes), `code/03_analysis.do` (estimates and figures), `code/04_make_tables.py` (tables and the numbers in this document), with logs in `logs/` and derived data in `data/` (`rd_analysis.dta` is the analysis file). Source folders under `Teachers Peru/Data` are treated as read-only.